

L36 — Bellman-Ford Algorithm

Unit: 3 | Chapter: Graphs | BT Level: BT5 | CO: CO3, CO4

Learning Objectives

- Explain why Bellman-Ford handles negative-weight edges where Dijkstra fails
 - Implement Bellman-Ford and trace its relaxation steps
 - Detect negative-weight cycles using Bellman-Ford
 - Compare Bellman-Ford and Dijkstra in terms of scope and complexity
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1. Motivation: Dijkstra's Limitation

Dijkstra's algorithm uses a greedy approach that relies on **non-negative edge weights**. When negative edges exist, Dijkstra can finalize a vertex too early with an incorrect distance.

Bellman-Ford handles negative-weight edges by relaxing all edges **V-1 times** (iteratively), guaranteeing convergence.

2. Key Idea

Observation: In a graph with V vertices, the shortest path between any two vertices (with no negative cycles) has at most **V-1 edges**.

Algorithm: Relax all edges V-1 times. After the k-th iteration, all shortest paths using at most k edges are correctly computed.

3. Algorithm

```
Algorithm BellmanFord(graph, source):
    dist[v] = ∞ for all v
    dist[source] = 0
    parent[v] = None for all v

    // V-1 relaxation rounds
    for i from 1 to V-1:
        for each edge (u, v, weight) in graph:
            if dist[u] + weight < dist[v]:
                dist[v] = dist[u] + weight
                parent[v] = u

    // Detect negative cycle: if any edge can still be relaxed, cycle exists
    for each edge (u, v, weight) in graph:
        if dist[u] + weight < dist[v]:
            raise NegativeCycleError("Negative cycle detected")
```

```
return dist, parent
```

4. Implementation

```
def bellman_ford(vertices, edges, source):
    """
    vertices: list of vertex names
    edges: list of (u, v, weight) tuples
    source: starting vertex

    Returns (dist, parent) or raises ValueError if negative cycle exist
    """
    dist = {v: float('inf') for v in vertices}
    dist[source] = 0
    parent = {v: None for v in vertices}

    V = len(vertices)

    # V-1 relaxation rounds
    for _ in range(V - 1):
        updated = False
        for u, v, w in edges:
            if dist[u] != float('inf') and dist[u] + w < dist[v]:
                dist[v] = dist[u] + w
                parent[v] = u
                updated = True
        if not updated:  # Early termination if no updates
            break

    # Check for negative cycles
    for u, v, w in edges:
        if dist[u] != float('inf') and dist[u] + w < dist[v]:
            raise ValueError(f"Negative weight cycle detected involving")

    return dist, parent


def reconstruct_path(parent, source, target):
    """Trace back from target to source using parent pointers."""
    path = []
    node = target
    while node is not None:
        path.append(node)
        node = parent[node]
        if node == target:  # Detect cycle in path (shouldn't happen i
            break
    path.reverse()
    if path[0] != source:
        return None  # Target not reachable
    return path
```

5. Detailed Trace

Graph:

A $\rightarrow$ (6) B
A $\rightarrow$ (7) C
B $\rightarrow$ (5) C B $\rightarrow$ (-4) D
C $\rightarrow$ (-3) B C $\rightarrow$ (9) D
D $\rightarrow$ (2) A D $\rightarrow$ (7) B

Vertices: {A, B, C, D, E}, Source: A

Edges: (A,B,6), (A,C,7), (B,C,5), (B,D,-4), (C,B,-3), (C,D,9), (D,A,2), (D,B,7)

Initial: `dist = {A:0, B:∞, C:∞, D:∞}`

Round 1 (relax all edges):

- (A,B,6): `dist[B] = 0 + 6 = 6` ✓
- (A,C,7): `dist[C] = 0 + 7 = 7` ✓
- (B,C,5): `dist[C] = min(7, 6 + 5 = 11) → no update`
- (B,D,-4): `dist[D] = 6 + (-4) = 2` ✓
- (C,B,-3): `dist[B] = min(6, 7 + (-3) = 4) = 4` ✓
- (C,D,9): `dist[D] = min(2, 7 + 9 = 16) → no update`
- (D,A,2): `dist[A] = min(0, 2 + 2 = 4) → no update`
- (D,B,7): `dist[B] = min(4, 2 + 7 = 9) → no update`

After Round 1: {A:0, B:4, C:7, D:2}

Round 2:

- (A,B,6): `dist[B] = min(4, 0 + 6) → no update`
- (B,C,5): `dist[C] = min(7, 4 + 5 = 9) → no update`
- (B,D,-4): `dist[D] = min(2, 4 - 4 = 0) = 0` ✓
- (C,B,-3): `dist[B] = min(4, 7 - 3 = 4) → no update`
- (D,B,7): `dist[B] = min(4, 0 + 7 = 7) → no update`

After Round 2: {A:0, B:4, C:7, D:0}

Round 3 — no updates: Algorithm converges.

Final distances from A: A=0, B=4, C=7, D=0

6. Negative Cycle Detection

```
def has_negative_cycle(vertices, edges, source):  
    """Return True if graph has a negative cycle reachable from source.  
    dist = {v: float('inf') for v in vertices}  
    dist[source] = 0  
  
    for _ in range(len(vertices) - 1):  
        for u, v, w in edges:
```

```

        if dist[u] != float('inf') and dist[u] + w < dist[v]:
            dist[v] = dist[u] + w

# V-th round: if any relaxation still possible → negative cycle
for u, v, w in edges:
    if dist[u] != float('inf') and dist[u] + w < dist[v]:
        return True

return False

```

Why does the V-th round detect cycles?

After V-1 rounds, all shortest paths are finalized (they have at most V-1 edges). If a V-th round still reduces a distance, it means there's a cycle of negative total weight that can be traversed indefinitely.

7. Bellman-Ford vs Dijkstra

Feature	Dijkstra	Bellman-Ford
Handles negative edges	No	Yes
Detects negative cycles	No	Yes
Time complexity	$O((V + E) \log V)$	$O(VE)$
Space complexity	$O(V)$	$O(V)$
Algorithm type	Greedy	Dynamic Programming
Implementation	Priority queue	Iterate all edges V-1 times
Use case	GPS routing (non-negative weights)	Financial arbitrage, general SSSP

8. Application: Currency Arbitrage

Currencies are vertices; exchange rates are edges.

Transform: **weight** = $-\log(\text{exchange_rate})$

Then a negative cycle = a sequence of exchanges that yields profit.

```

import math

# Exchange rates
rates = {
    ('USD', 'EUR'): 0.85,
    ('EUR', 'GBP'): 0.90,
    ('GBP', 'USD'): 1.35,
}

# Convert to negative log
log_edges = [(u, v, -math.log(r)) for (u, v), r in rates.items()]

# If Bellman-Ford detects a negative cycle → arbitrage opportunity

```

Summary

- Bellman-Ford: relaxes all edges $V-1$ times $\rightarrow O(VE)$ time.
- Handles **negative-weight edges** (Dijkstra cannot).
- Detects **negative-weight cycles** in a V -th relaxation round.
- Correctness: after k rounds, all shortest paths with $\leq k$ edges are correct.
- Use Dijkstra for efficiency when weights are non-negative; Bellman-Ford when negative edges or cycle detection is needed.

References

- Cormen et al., *Introduction to Algorithms*, Ch. 24.1 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 8.2 (Springer, 2020)